UNIVERSIDAD NACIONAL DE COLOMBIA

Respuestas - Taller de Calculo Integral II

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1 Integrales Definidas y Propiedades

1. $\int_{-2}^{2}(x^5+3x^2)\,dx = 16$.
2. $\int_{0}^{3}|x-1|\,dx = \frac{5}{2}$.
3. $\int_{1}^{4}(2x-3)\,dx = 6$.
4. $\int_{0}^{\pi}\cos x\,dx = 0$.
5. $\int_{0}^{6}f(x)\,dx = 5+(-3) = 2$.

2 Teorema Fundamental del Calculo

1. Si $F(x) = \int_{1}^{x}(t^2+2t)\,dt$, entonces

$$F'(x) = x^2+2x, \qquad F(3) = \int_{1}^{3}(t^2+2t)\,dt = \frac{50}{3}.$$

2. Si $G(x) = \int_{x}^{2x}\ln(1+t^2)\,dt$, entonces

$$G'(x) = 2\ln(1+4x^2) - \ln(1+x^2).$$

3.

$$\frac{d}{dx}\left(\int_{0}^{x^3}\sqrt{1+t^4}\,dt\right) = \sqrt{1+x^{12}}\cdot 3x^2.$$

4. Para $H(x) = \int_{\sin x}^{\cos x}(1+t^3)\,dt$,

$$H'(x) = -(1+\cos^3 x)\sin x - (1+\sin^3 x)\cos x.$$

3 Area Entre Curvas

1. Entre $y = x$ y $y = x^2$:

$$A = \int_{0}^{1}(x-x^2)\,dx = \frac{1}{6}.$$

2. Entre $y = 4 - x^2$ y eje x:

$$A = \int_{-2}^{2} (4 - x^2)\, dx = \frac{32}{3}.$$

3. Entre $y = 2x + 3$ y $y = x^2$ (intersecciones en $x = -1, 3$):

$$A = \int_{-1}^{3} (2x + 3 - x^2)\, dx = \frac{32}{3}.$$

4. Entre $y = \sin x$ y $y = \cos x$ en $[0, \pi/2]$:

$$A = \int_{0}^{\pi/4} (\cos x - \sin x)\, dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\, dx = 2(\sqrt{2} - 1).$$

4 Volumenes por Discos y Arandelas

1. Giro de $y = x^2$, $0 \leq x \leq 2$, alrededor de eje x:

$$V = \pi \int_{0}^{2} (x^2)^2\, dx = \frac{32\pi}{5}.$$

2. Giro de $y = \sqrt{x}$, $0 \leq x \leq 4$, alrededor de eje x:

$$V = \pi \int_{0}^{4} (\sqrt{x})^2\, dx = \pi \int_{0}^{4} x\, dx = 8\pi.$$

3. Giro de la region bajo $y = 4 - x$ en torno al eje y:

$$V = \pi \int_{0}^{4} (4 - y)^2\, dy = \frac{64\pi}{3}.$$

5 Longitud de Arco y Promedio

1. Para $y = \frac{x^2}{2}$ en $[0, 1]$, $y' = x$:

$$L = \int_{0}^{1} \sqrt{1 + x^2}\, dx = \frac{1}{2}\left(\sqrt{2} + \ln(1 + \sqrt{2})\right).$$

2. Valor promedio de $f(x) = x^2 + 1$ en $[0, 3]$:

$$f_{\text{prom}} = \frac{1}{3} \int_{0}^{3} (x^2 + 1)\, dx = 4.$$

3. Valor promedio de $f(x) = \sin x$ en $[0, \pi]$:

$$f_{\text{prom}} = \frac{1}{\pi} \int_0^{\pi} \sin x \, dx = \frac{2}{\pi}.$$